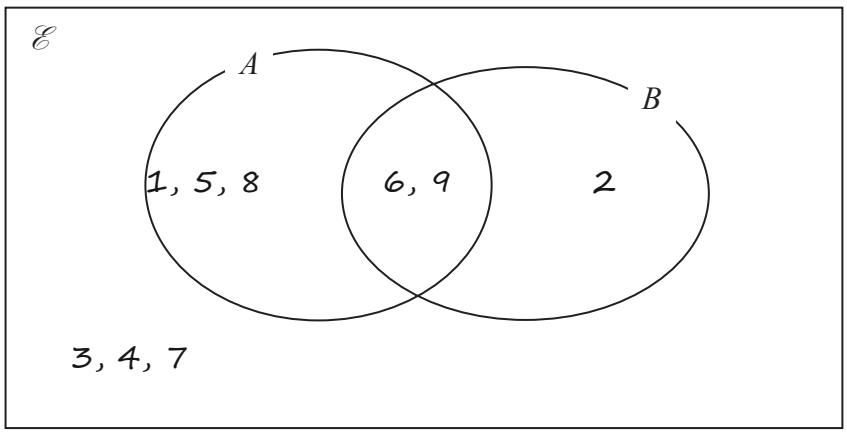
1 $\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$
$A = \{1, 5, 6, 8, 9\}$
$B = \{2, 6, 9\}$



(a) Complete the Venn diagram to represent this information.

(3)

A number is chosen at random from the universal set $\mathscr{E}$.

(b) Find the probability that the number is in the set $A \cap B$

$A \cap B = \{6, 9\}$: 2 outcomes from 9.

$P(A \cap B) = 2/9$

1 mark
Count 2 of 9 outcomes.

1 mark
Overlap {6,9}.

1 mark
Place remaining values.

1 mark
All nine values placed.

1 mark
Probability 2/9.

2/9

(2)

(Total for Question 1 is 5 marks)